

Mansi Tanna

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PROFESSIONAL SUMMARY

Graduate student in Data Analytics with strong foundations in computer science, systems programming, and software engineering. Experience designing scalable distributed systems & real-time data pipelines with emphasis on reliability, modular architecture, debugging, and measurable impact

EDUCATION

San Jose State University

Master of Science, Data Analytics

Aug 2024 - May 2026

Charotar University of Science & Technology

Bachelor of Technology, Computer Science

Aug 2020 - May 2024

SKILLS

- **Languages & Frameworks:** Python, JavaScript/TypeScript, C/C++, Node.js, Express.js, React.js, Redux, MERN Stack, PHP, Git
- **Backend & Cloud:** RESTful APIs, AWS, Docker, Kubernetes, Kafka, Redis, MySQL, MongoDB, Snowflake, Apache Airflow
- **Software Engineering & DevOps:** CI/CD, GitHub Actions, Jenkins, microservices architecture, Agile/Scrum, code reviews, PyTest, Jest, Debugging, Automation, Secure Coding, Container Orchestration, Monitoring & Logging
- **AI & Data Systems:** TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, Feature Engineering, Predictive Analytics, Generative AI
- **Tools:** dbt, Docker, Git, VS Code, PyCharm, IBM Watsonx Orchestrate, Watson AI Tools, Agile Collaboration

EXPERIENCE

Research Assistant — Dept. of Computer Engineering, San Jose State University

May 2025 - Dec 2025

- Designed & evaluated an in-vehicle audio pipeline using ESC-50 and AudioSet to support real-time environmental & safety-aware decision systems
- Benchmarked CLAP and CNN-LSTM models across **1,000+ clips**, streamlining preprocessing and reducing training cycles by **40%**
- Integrated a **reinforcement learning-based decision layer (DQN, PPO)** to improve adaptive system behavior and overall model robustness
- Contributed structured documentation of evaluation metrics and system performance for research publication

Tech: Python, TensorFlow, Keras, NumPy, Pandas, Docker, GitHub Actions

Research Intern — Indian Space Research Organisation (ISRO)

Jan 2024 - Jun 2024

- Developed a deep learning-based image processing system to support **earlier identification of fire-prone regions** using multispectral satellite imagery
- Achieved 88% detection accuracy on **500+ real-world satellite images** through targeted CNN tuning and controlled experimentation
- Automated spectral band normalization and noise reduction pipelines to enhance reliability and inference consistency
- Collaborated with scientists to validate results and ensure operational robustness of the system

Tech: TensorFlow, OpenCV, Python, CUDA, AWS EC2

Software & Data Engineering Intern — BrainyBeams Technologies Pvt. Ltd

May 2023 - Aug 2023

- Built scalable ETL microservices to process **100K+ sales records**, giving business teams visibility into revenue trends and operational patterns
- Boosted forecasting accuracy by **18%** through regression modeling and deployed updates via **automated CI/CD pipelines** for stable, repeatable releases

Tech: Python, Pandas, NumPy, Docker, GitHub Actions, Power BI, Superset, MySQL

Web Development Intern — PHP Dots

May 2022 - Aug 2022

- Developed responsive front-end interfaces, refining user experience and decreasing page load time by **30%**

Tech: HTML, CSS, JavaScript, Bootstrap, PHP

PROJECT EXPERIENCE

Uber System Simulation

- Designed a **distributed ride-sharing simulation** supporting **1,000+ concurrent users**, deploying **Kafka-based microservices** and real-time asynchronous communication
- Lowered latency by **25%** through performance tuning on AWS and optimized message coordination across services

Tech: Node.js, Express.js, MySQL, MongoDB, Kafka, Redis, Docker, Kubernetes, AWS, JMeter

Health Risk Prediction

- Configured a **Spark Streaming pipeline** processing **10GB+ wearable IoT data** to support **early detection of high-risk cardiovascular events**
- Raised model accuracy by **10%** via feature selection and parallelized Spark processing to improve system responsiveness

Tech: Spark, Hadoop, Kafka, Python, Scikit-Learn, Pandas, AWS

ML-Powered Shortlisting & Interview Aid

- Created an NLP-based resume shortlisting system with **SHAP explainability** and GPT-4 automation, reducing **manual recruiter screening by 40%** while improving transparency and consistency

Tech: Python, Scikit-Learn, GPT-4 API, Flask, SHAP, Pandas, NumPy

Graph-Based Few-Shot Threat Detection

- Built a scalable threat detection system on **large-scale network datasets** to detect **rare cyber attacks missed by baseline ML and LLM-only models**
- Outperformed baseline classifiers by **~10%** on rare-threat detection using graph-based learning and retrieval grounding

Tech: Python, Spark, Airflow, GraphFrames, NetworkX, FAISS, GNNs, RAG

HACKATHONS & PROGRAMS

SJSU × IBM SkillsBuild Hackathon

Selected for the SJSU × IBM SkillsBuild Hackathon; collaborated with a new team to design SpartanCare, an AI health navigator using IBM Watsonx Orchestrate